MAE Task : Motocross Jump Analytics
Technical report by Sachin Sagar

1. Setup & Dataset Specification:

- 6 axis IMU + gyro
- Sampling Rate : 500 Hz
- time-frame : 90s
- frame of reference :RHS with X = forward, Y = left, Z = up
- IMU mounted on the frame of a bike with 2-stroke engine

2. Additional Observations from pre-processing:

- a 0.4s sensor dropout (all signals) at t = 23s
- faulty pitch-rate saturation to full scale reading of 8 rad/s during the 4th jump
- mean yaw-rate slowly drifts after the 4th jump

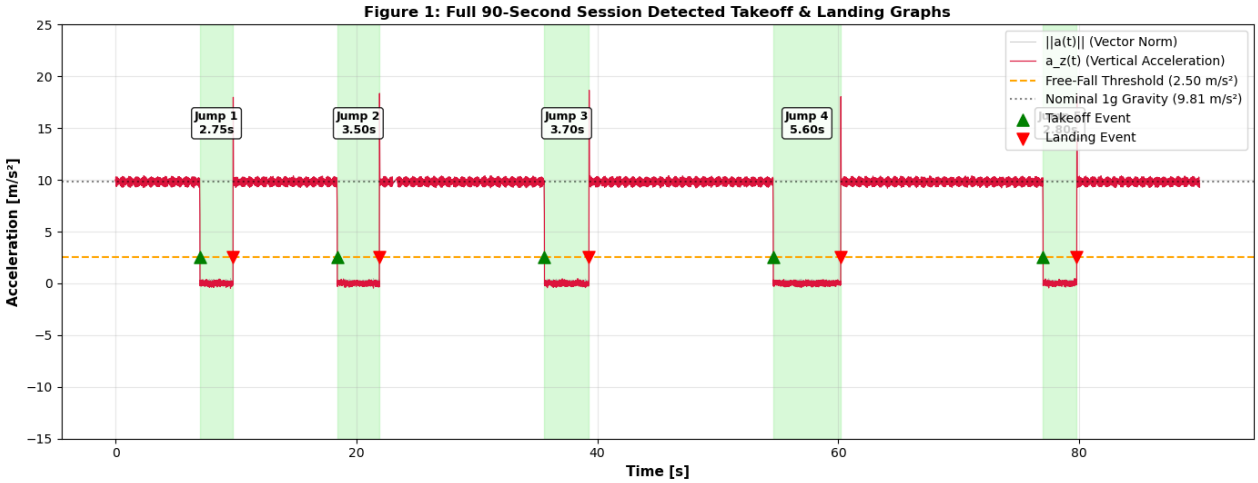
A. Objective 1: Jump Detection

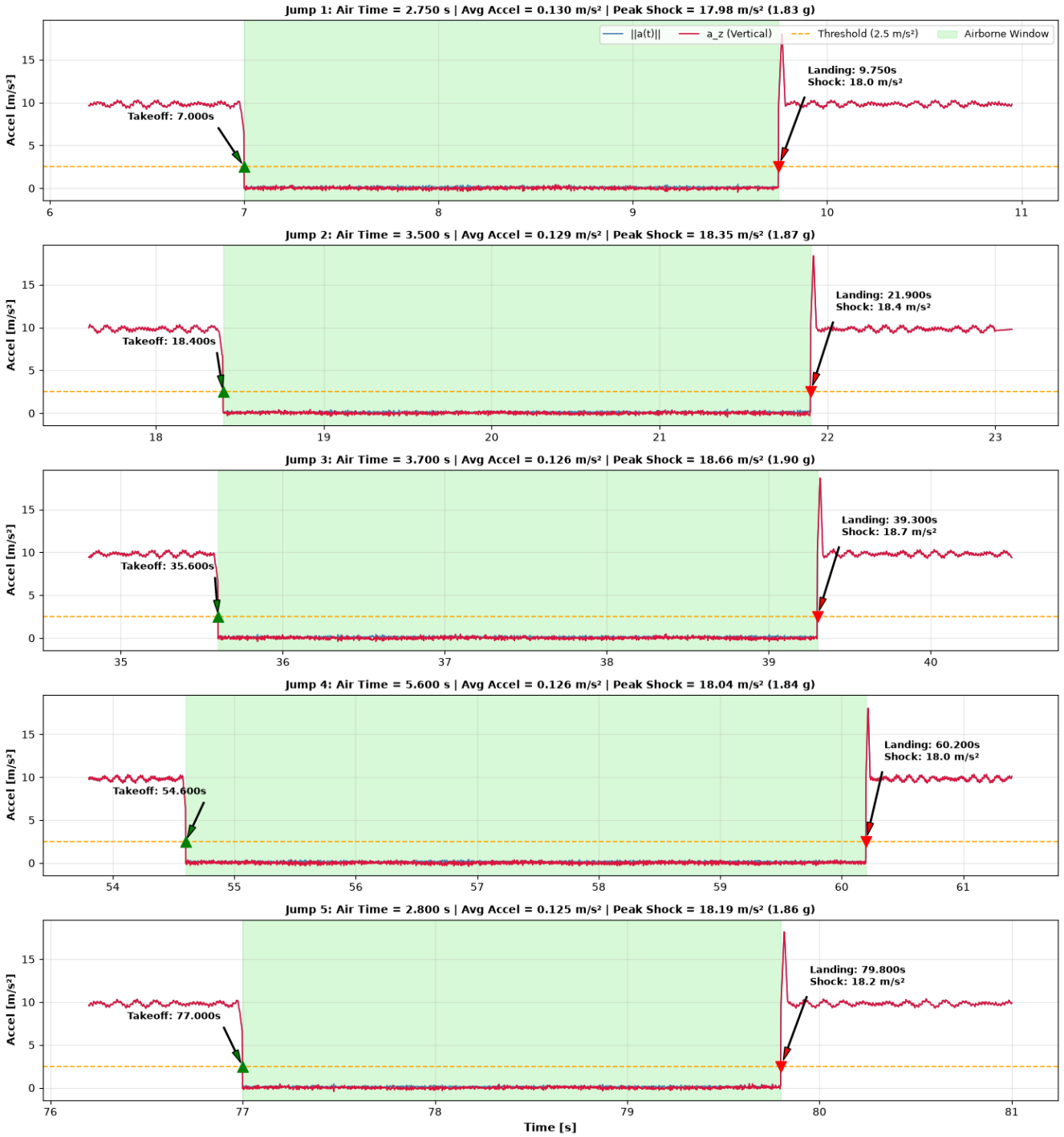
1. Detection Criteria

- (**L2 norm of acceleration vector < a_thrs**) for (**del_t > tau_thrs**) and a success counter **K_conf iterations**
- Reasoning :
  - When bike is in flight, the the ground reaction vanishes and the IMU observes 0 net acceleration
  - A thresholding is needed to account for sensor noise and a debounce and success counter are needed to suppress false positives from short lived suspension dynamics.

2. Graphs :

- **a_thrs = 2.5 mps2 t_thrs = 0.5 s (250 samples @500Hz) K_conf = 2**
- The takeoff and landing events are identified using the step change in net acceleration





3. Air-time stats:

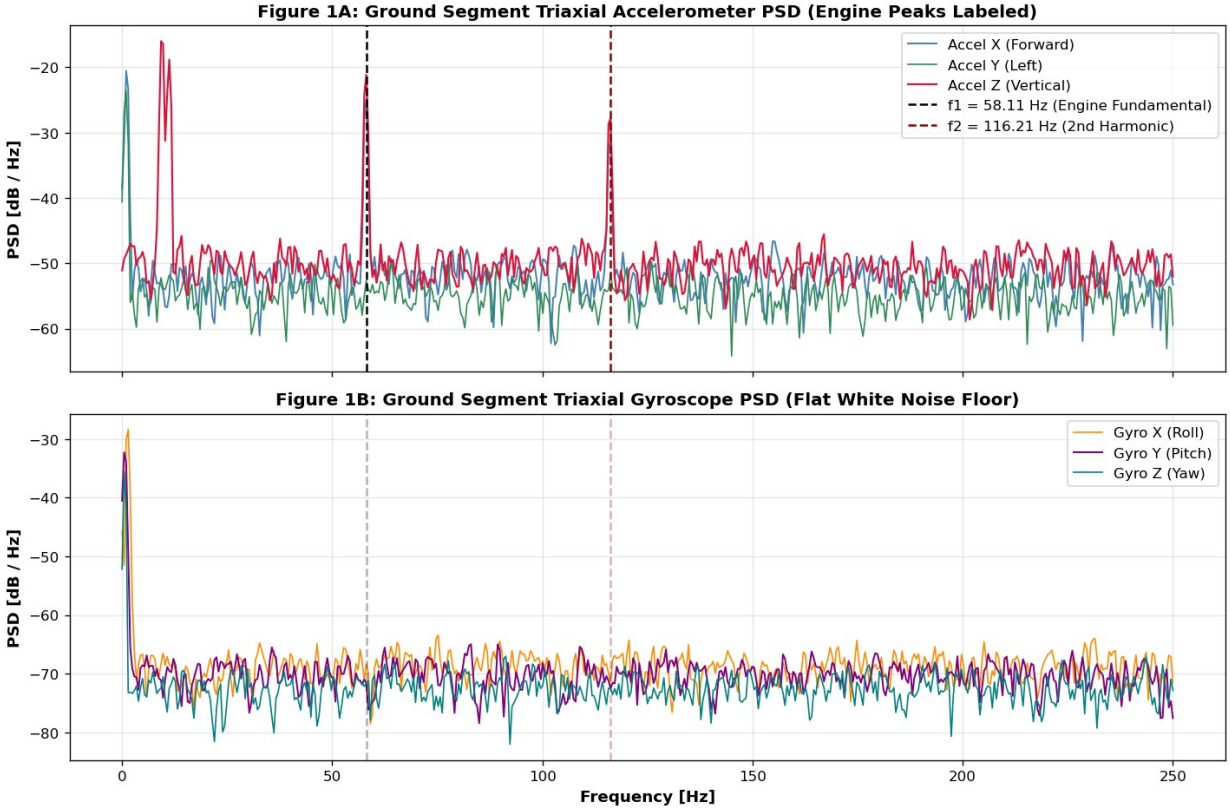
Jump Idx	Takeoff time (s)	Landing time (s)	Air time (s)	Airborne a_net (mps2)	Landing a_net (mps2)
1	7.0	9.75	2.75	0.13	17.98
2	18.4	21.9	3.5	0.129	18.35
3	35.6	39.3	3.7	0.126	18.66
4	54.6	60.2	5.6	0.126	18.04
5	77.0	79.8	2.8	0.125	18.19

B. Objective 2: Filter Design

- **The two stroke engine produces strong vibrations at:**
  - **fundamental ~ 58 Hz**
  - **secondary ~ 116Hz**

1. PSD for acceleration and gyro during ground segment [11.0, 17.0] s

- for triaxial acceleration and gyro rate :



- Identified peaks:
  - **fundamental = 58.11 Hz**
  - **secondary = 116.21 Hz**
- These peaks are only observed in x and z accelerometer readings
- The gyro rates are unaffected by engine primary and secondary harmonics
- The **motion dynamics** appear concentrated around **20 Hz**

2. Butterworth filter design

- **constraint : preserve motion signal below 20Hz**

Filter Design:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}}$$

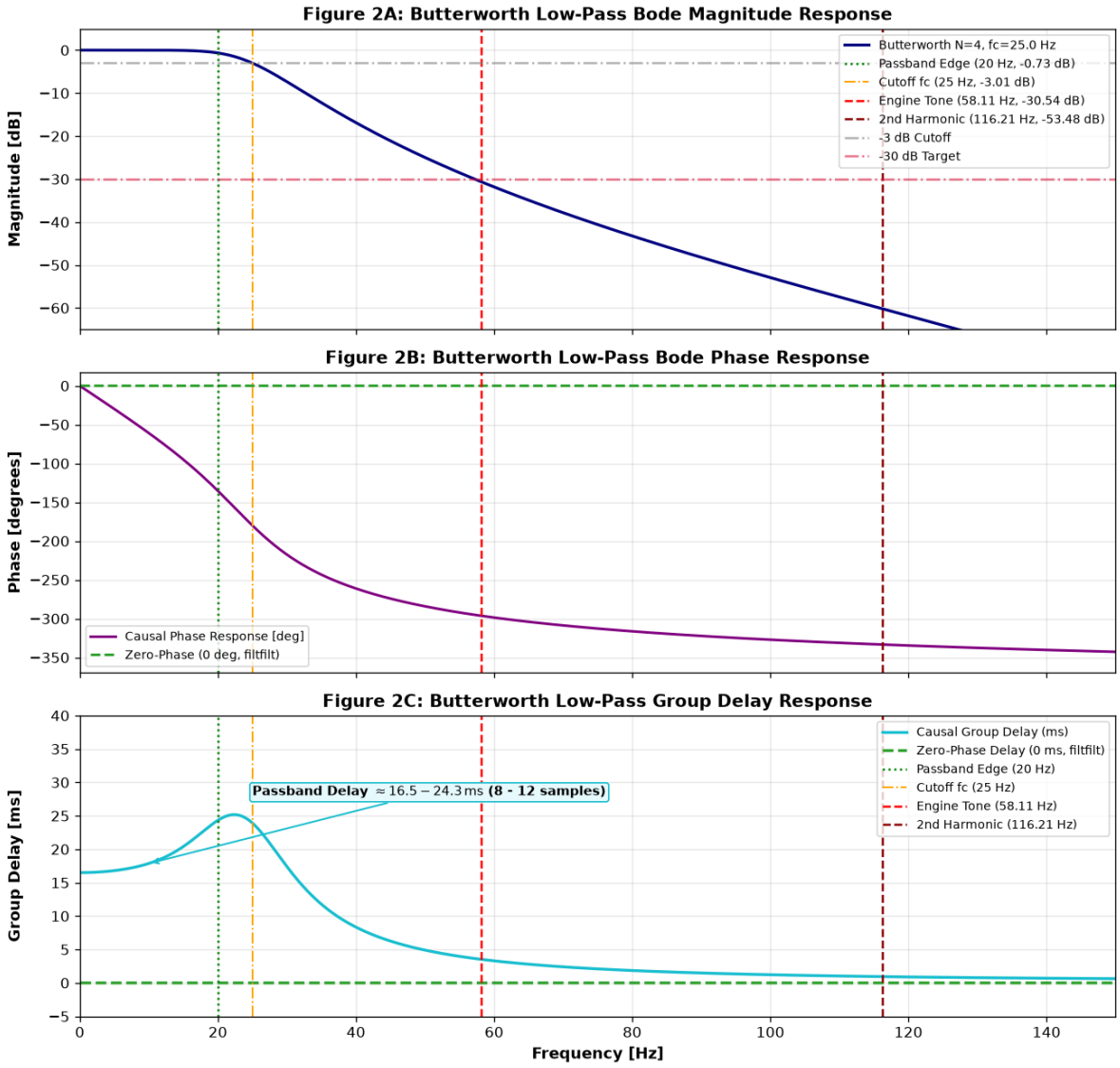
$$\left(\frac{f_s}{f_p}\right)^{2N} \geq \frac{10^{A_s/10} - 1}{10^{A_p/10} - 1}$$

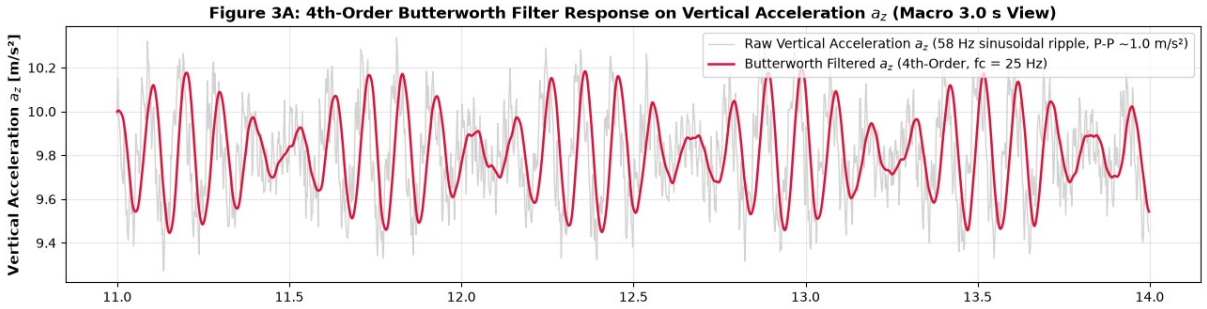
- pass-band : **f_p = 20Hz** with max attenuation **A_p = 1 dB**
- stop-band : **f_s = 58Hz** with min attenuation **A_s = 30dB**

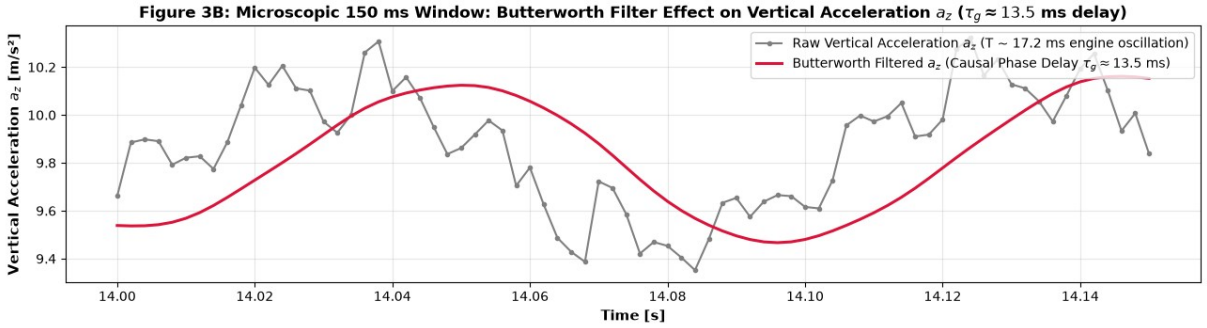
- Required filter Order **N > 3.88**
- Hence **N = 4**
- setting half power **cutoff** at **-3 dB** we get cut-off frequency **f_c = 25Hz**

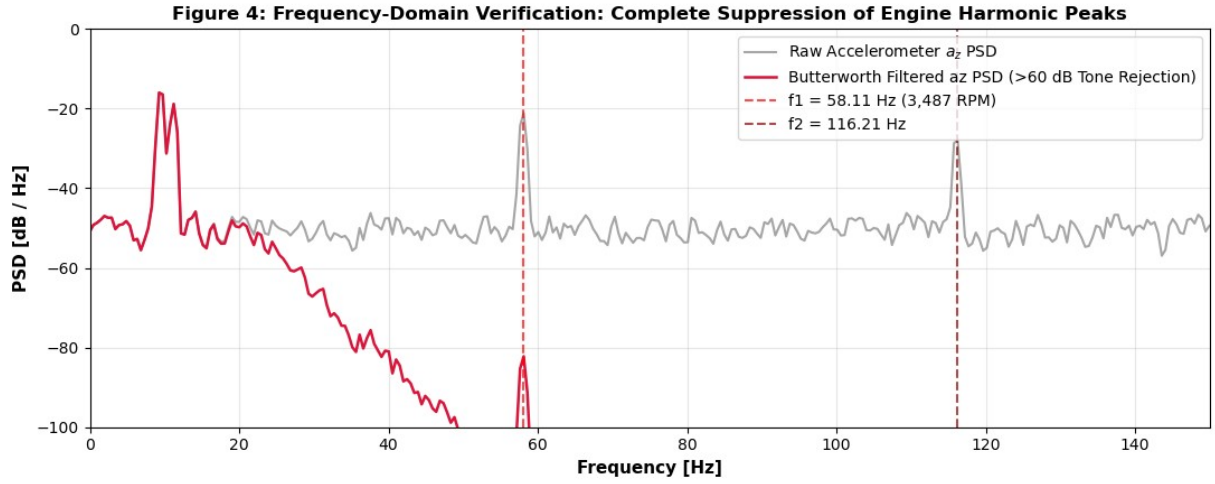
- The Bilinear Transform is used to discretize the filter (s → z domain)
- A product of two bi-quad filters is used instead of a single 4th order transfer function for numerical stability
- The coefficients are computed numerically using the signal library in scipy

Filter Response:





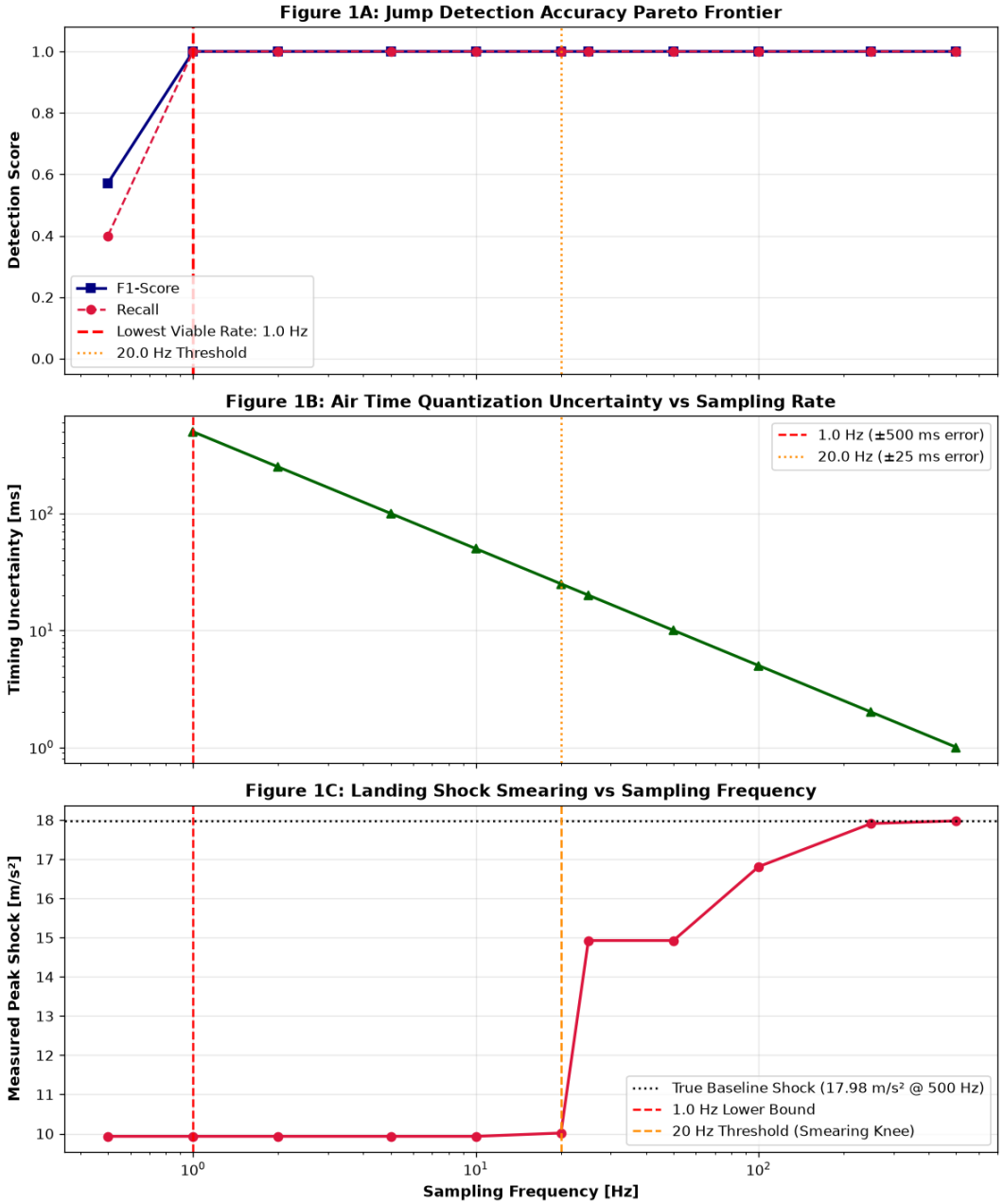




C. Objective 3: Down-sampling and power efficiency

1. Lowest sample rate for jump detection

- smallest jump tenure in the dataset del(t_min) = 2.75 s
- To prevent chatter from pot-holes and speed bumps, we implement a confirmation counter (K_conf) of 2.
- **f_crit > K_conf/del(t_min) = 0.73 Hz**
- hence, we set **f_crit = 1 Hz**
- The binary jump detection remains **unaffected** by the severe down-sampling till f_crit of 1 Hz.

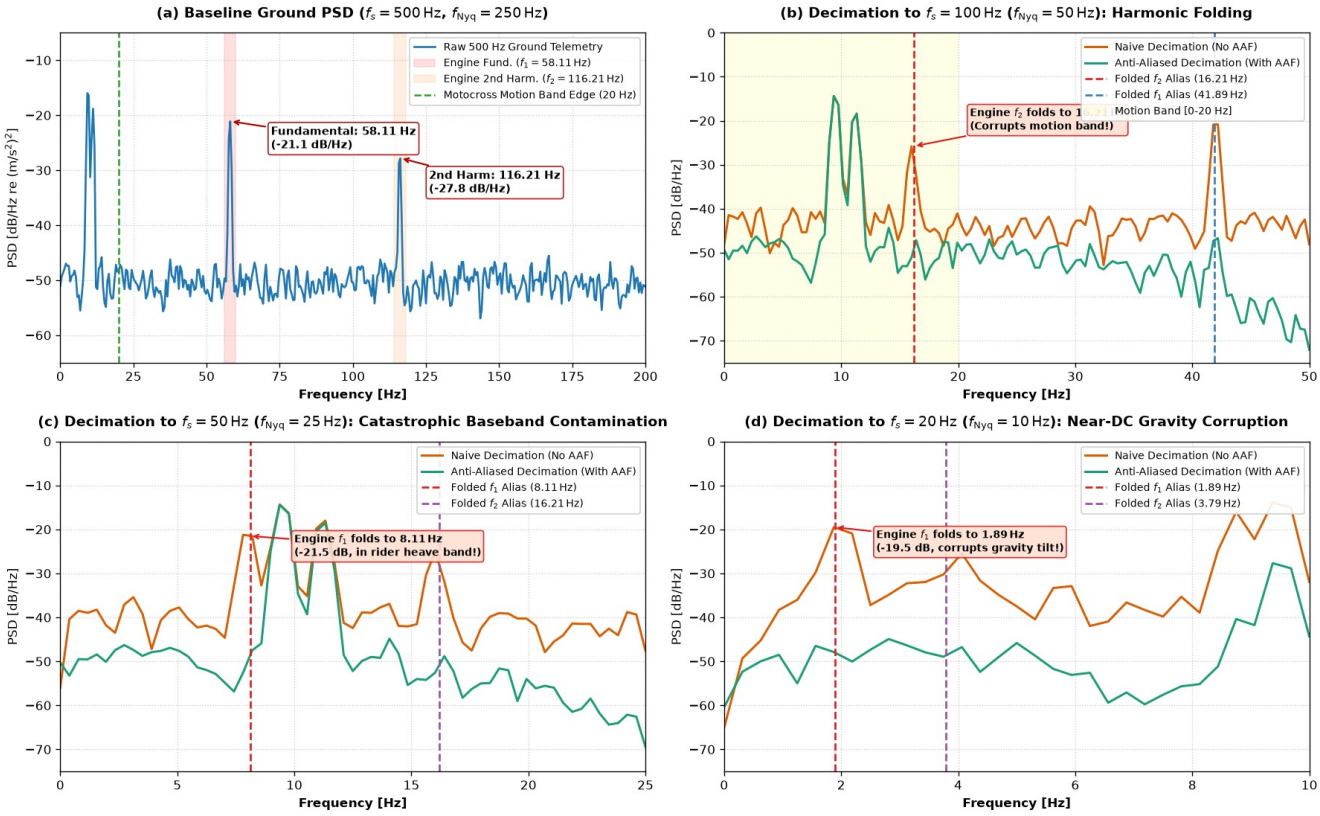


- However, related parameters like air time acquire a significant error and shock magnitude become unobservable

2. Comment on Antialiasing filters at lower frequencies:

- Antialiasing filters (AAF) are not needed for the binary jump detection till a decimation of 500 (1Hz sampling)
- The Engine noise folds over the motion dynamics band for sampling rates < 100Hz
- If the motion dynamics are to be preserved, a minimum sampling rate of 50 Hz is needed (Nyquist limit for f_c = 25Hz), with an AAF

Figure 2: Spectral Aliasing & Nyquist Folding Analysis of Motocross Engine Harmonics
Empirical Verification of $f_{alias} = |f_0 - k \cdot f_s|$ and Essential Anti-Aliasing Filter (AAF) Suppression



Conclusion:

1. 5 jumps detected, 18.32s total air time, zero false alarms.

2. 4th order Butterworth filter (f_c = 25Hz) is used to remove engine harmonics

3. 1Hz minimal rate for binary detection, 50Hz for shock dynamics with mandatory AAF for f<100 Hz